

Neutrophil to Lymphocyte Ratio as a Novel Predictor of Outcome in Patients With Severe Traumatic Brain Injury

Weiqliang Chen, MD, PhD; Jinhua Yang, MD; Bingbing Li, MD; Guoyi Peng, MD; Tianfei Li, MD; Lianjie Li, MD; Shousen Wang, MD, PhD

Objective: The neutrophil to lymphocyte ratio (NLR) has been reported to be a predictor of outcome in critical illness. Objective of this study was to investigate the changes of the NLR in patients with severe traumatic brain injury (TBI) and analyze the relationship between the NLR and TBI outcome. **Setting:** China. **Participants:** A total of 855 patients with severe TBI from January 2007 to April 2012. **Design:** Retrospective. **Main Measurement:** Data on the NLR and other indicators were collected. After follow-up until death or 1 year, the relationship between the NLR and TBI outcome was analyzed retrospectively. **Results:** The final analysis included 688 patients. There were 508 (73.8%) who had an unfavorable outcome by 1 year after head trauma. The value of the NLR on admission was significantly higher in the unfavorable outcome group than in the favorable outcome group ($P < .001$). Multivariate logistic analysis showed that higher NLR was associated with an unfavorable outcome (odds ratio, 1.100; $P < .001$). Receiver operating characteristic curve analysis showed that the NLR had a sensitivity of 60.2% and a specificity of 71.1% for predicting unfavorable outcome at 1 year on the basis of the best threshold. **Conclusion:** The NLR might be useful as a novel predictor for 1-year outcome and mortality in severe TBI. **Key words:** Glasgow Coma Scale, Glasgow Outcome Scale, neutrophil to lymphocyte ratio, prognosis, traumatic brain injury

DESPITE EXTENSIVE study and improvements in critical care, outcomes after traumatic brain injury (TBI) continue to be poor and difficult to predict.

Author Affiliations: Department of Neurosurgery, Fuzhou General Hospital, Xiamen University Medical College, Fuzhou, China (Drs Chen, B. Li, T. Li, L. Li, and Wang); and Department of Neurosurgery, First Affiliated Hospital, Shantou University Medical College, Shantou, Guangdong, China (Drs Yang and Peng).

Drs Chen, Yang, and B. Li contributed equally to this work.

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Corresponding Author: Shousen Wang, MD, PhD, Department of Neurosurgery, Fuzhou General Hospital, Xiamen University Medical College, Fuzhou 350025, China (wshsen@xmu.edu.cn).

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In developed countries severe TBI remains the commonest cause of death in individuals younger than 40 years, and in developing countries the incidence and societal costs of TBI are rising.¹ There is a large body of evidence suggesting that neuroinflammation is an important injury mechanism that contributes to ongoing neurodegeneration and neurological impairments associated with TBI. Posttraumatic neuroinflammation is characterized by glial cell activation, leukocyte recruitment, and upregulation of inflammatory mediators.^{2–7} Although many studies have focused on the detrimental effects of neuroinflammation on the injured brain, there are clear beneficial effects that can be achieved if neuroinflammation is controlled by anti-inflammatory treatments for TBI. It has been shown that minocycline has significant neuroprotective effects in spinal cord injury models by reducing both inflammatory and apoptotic pathways.⁸ The peroxisome proliferator-activated receptor 3 (PPAR3), pioglitazone and rosiglitazone, reduced posttraumatic microglial activation and promoted antioxidant (eg, Mn-SOD) and neuroprotective chaperone protein (eg, HSP27/70) expression that resulted in improved functional and histological outcomes after TBI.^{9–11}

The Glasgow Coma Scale (GCS) is well established as a predictor of outcome in TBI. Osler et al⁸ found that a low GCS score is a better predictor of mortality

in patients with TBI than in patients without TBI. However, increasing research indicates that prognosis of patients with TBI is based not only on the GCS but also on host-related factors such as mild hypothermia,¹² Cushing ulcer,¹³ neuroinflammation,¹⁴ and system inflammation. In patients with mild TBI, it was reported that the presence of brain contusions and the size of foci of hemorrhagic axonal injury on MRI are predictive of outcome at 3 months.¹⁵ A moving correlation coefficient, called the PRx, which is based on cerebrovascular reactivity, is a predictor of outcome in TBI.¹⁶ It has also been reported that plasma high-mobility group box 1 levels, a system inflammation maker, predict outcome at 1 year in TBI,¹⁷ and in patients with severe TBI, blood mitochondrial enzymes have been found to predict long-term outcome.¹⁷ There is particularly a lot of evidence indicating that the systemic inflammatory response can predict outcome in TBI.

Inflammation plays a role in neurological diseases and neurological trauma. In intracerebral hemorrhage, activation of an inflammatory response as manifested by leukocytosis may affect the severity of the condition.¹⁸ Evidence is accumulating that there is a relationship between inflammation and ischemic stroke.^{19,20} There is also increasing evidence from studies using animal models that an inflammatory response during convulsive status epilepticus plays a role in seizures that occur and their long-term negative effects.²¹

The neutrophil to leukocyte ratio (NLR), which is a significant indicator for predicting the inflammatory status of patients, has been shown to be a predictor of prognosis in patients with conditions involving the brain including glial tumor,^{22,23} ischemic stroke,^{20,24} hemorrhagic stroke,²⁵ and convulsive status epilepticus.²¹ However, there has been a lack of research to investigate whether the NLR can predict outcome in patients with TBI. Therefore, the aim of this study was to assess the value of the NLR for predicting outcome in patients with TBI.

METHODS

Study population

This retrospective study was conducted from January 2007 to April 2012. The inclusion criteria were (1) isolated head trauma, (2) a posttrauma GCS score 8 or less, and (3) time from injury to admission 6 hours or less. The exclusion criteria were (1) age less than 18 years, (2) time from injury to admission time more than 6 hours, (3) previous head trauma, (4) neurological disease including ischemic or hemorrhagic stroke, (5) use of antiplatelet or anticoagulant medication, (6) use of steroids or immunosuppressant, and (7) presence of prior systemic diseases including uremia, liver cirrhosis, malig-

nancy, chronic heart or lung disease, diabetes mellitus, and hypertension.²⁵

This study was approved by the ethics committees of both participating hospitals (first affiliated hospital of Shantou University Medical College and Fuzhou General Hospital of Xiamen University Medical College). Informed consent for each patient in the study was obtained from the patients or their relatives.

Clinical and radiological assessment

Clinical data were obtained by reviewing medical charts. Severity of TBI was assessed on the basis of the initial postresuscitation GCS score after admission. Shock was defined as a systolic blood pressure of less than 90 mm Hg, and hypoxia was defined as blood oxygen saturation less than 85%.²⁰ Hyperglycemia was defined as blood glucose more than 11.1 mmol/L,²¹ and hypoglycemia was defined as blood glucose less than 2.2 mmol/L.²⁶ Neurological deterioration was defined as occurring in patients who manifested clinically identified episodes of 1 or more of the following: (1) a spontaneous decrease in GCS motor scores of 2 points or more from the previous examination, (2) a further loss of pupillary reactivity, (3) development of pupillary asymmetry more than 1 mm, and (4) deterioration in neurological status sufficient to warrant immediate medical or surgical intervention.²⁰ All computerized tomography (CT) scans were performed according to the neuroradiology department protocol. Investigators who read the scans were blinded to clinical information. CT classification was performed using Traumatic Coma Data Bank criteria on initial postresuscitation CT scan.²³

Patient management

Treatments included surgical therapy, ventilatory support, arterial pressure maintenance, glycemic control, intravenous fluids, hyperosmolar agents, H₂ blockers, early nutritional support, and physical therapy. The decision to perform tracheal intubation and use of mechanical ventilation was based on the patient's level of consciousness, and the ability to protect the airway and maintain arterial blood gas concentrations. Achieving adequate intravascular volume was pursued aggressively; vasopressors are used only after volume expansion. When clinical and radiological examinations provided an estimate of increased intracranial pressure, osmotherapy in the form of intravenous mannitol was administered, if available, deepening sedation and hyperventilation. Intracranial mass lesions associated with midline displacement more than 5 mm were surgically removed when necessary. If intracranial pressure remained high after maximal medical therapy was performed or an intracranial mass lesion was removed, decompressive craniotomy was performed as soon as possible.²⁴

Collection of index data

Venous blood was withdrawn on admission. The index data obtained included age; GCS score on admission; time from injury to admission time (hour); plasma-sampling time (hour); systolic, and diastolic blood pressure (mm Hg); mean arterial pressure (mm Hg); heart rate (beats/min); body temperature ($^{\circ}\text{C}$); blood oxygen saturation (percentage); blood glucose level (mmol/L); plasma C-reactive protein level (mg/L); sex; shock on admission; hyperglycemia on admission; hypoglycemia on admission; hypoxia on admission; pupil bluntness on admission; CT classification 5 or 6; abnormal cisterns on initial CT scan; midline shift more than 5 mm on initial CT scan; presence of traumatic subarachnoid hemorrhage on initial CT scan; neurological deterioration; mechanical ventilation; intracranial surgery during the initial 24 hours; and seizure. The NLR was calculated as neutrophil count divided by lymphocyte count. The blood samples were run in duplicate. Researchers who collected data and dealt with data were blinded to all patient details.

End point

If the outcome was not recorded in the medical chart, telephone interviews were performed by 1 doctor who was blinded to clinical information and the NLR to obtain the follow-up information. Patients were followed up until death or until 1 year after head trauma. Functional outcome was based on the Glasgow Outcome Scale (GOS) score. The GOS was defined as 1 = death, 2 = persistent vegetative state, 3 = severe disability, 4 = moderate disability, and 5 = good recovery.²⁵ GOS scores were divided into favorable (GOS 4–5) and unfavorable (GOS 1–3) outcomes, and in a similar way, into death (GOS 1) and survival (GOS 2–5).¹¹ The causes of death were not included in the analysis because except for the patients who died in the hospital the possible connection between TBI and death could not be established with certainty on the basis of reports from the families.

Statistical analysis

Statistical analysis was performed with SPSS 13.0 (SPSS Inc, Chicago, Illinois) and MedCalc 9.6.4.0 (MedCalc Software, Mariakerke, Belgium). Continuous variables were expressed as mean $\pm$ standard deviation and analyzed by the 2-sample *t* test. Categorical variables were expressed as counts (percentage) and analyzed by using the Pearson χ^2 test or Fisher exact test. A multivariate logistic regression model analysis was performed for identifying the association of 1-year (unfavorable) outcome and mortality with predictors (selection: forward [method = Wald]). Results were presented as odds

ratio (OR), 95% confidence intervals (95% CIs), and corresponding model summary statistics (overall model significance level, Cox and Snell Pseudo- R^2 , and Nagelkerke Pseudo- R^2) of the model selections.⁹ Receiver operating characteristic (ROC) curve analysis was also performed to evaluate the performance of GCS scores and NLR values on admission. Results were represented as area under the ROC curve (AUC) with 95% CI. The corresponding sensitivity and specificity were also calculated on the basis of the best threshold either for GCS scores and NLR values on admission derived from the ROC curve analysis. All statistical assessments were 2-tailed and significance was set at $P < .05$.

RESULTS

Patient characteristics

During the study period, 855 patients with a diagnosis of isolated head trauma were admitted to our intensive care unit. Of these patients, 94 were excluded for the reasons shown in Figure 1. A total of 688 were included in the analysis. Patient demographics, clinical characteristics, CT scan results, medication treatment, and laboratory examination are compared by unfavorable outcome and survival status in Table 1. Analysis of laboratory results showed that the NLR was associated with 1-year unfavorable outcome and mortality (both $P < .05$).

Functional outcome prediction

There were 508 patients (73.8%) who had an unfavorable outcome at 1 year after head trauma. Three variables—age, temperature, and the NLR on admission—that were significantly associated with 1-year unfavorable outcome were put into multivariate model analysis. The results showed that age and the NLR on admission were retained as significant predictors associated with 1-year unfavorable outcome (age: OR = 1.068, 95% CI = 1.052–1.083; NLR: OR = 1.100, 95% CI =

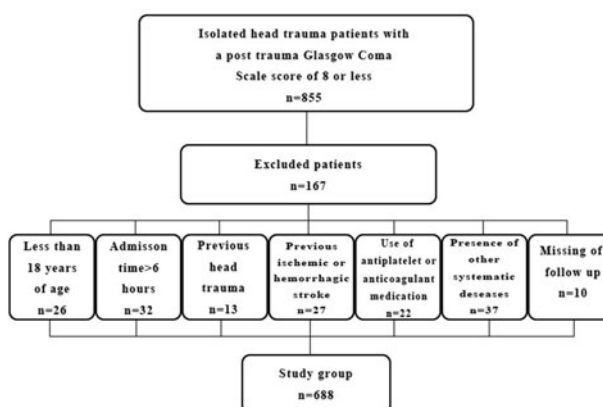


Figure 1. Flow chart showing how exclusion criteria were applied to select patients for the study.

TABLE 1 Patients' demographics, clinical characteristics, CT scan results, medication treatment, and laboratory examination by unfavorable outcome or survival status^a

Characteristics	Total	Unfavorable outcome		Survival status	
		Unfavorable	Favorable	Dead	Alive
Demographics					
Sex, male/female	557/131	413/95	144/36	212/36	345/95
Age, y	45.40 ± 14.85	48.58 ± 14.34	36.43 ± 12.42	44.74 ± 14.20	45.77 ± 15.21
Clinical characteristics					
GCS score on admission	5.95 ± 1.69	5.62 ± 1.67	6.90 ± 1.33	4.83 ± 1.55	6.60 ± 1.40
Shock on admission	180 (26.2%)	132 (26.0%)	48 (26.7%)	78 (31.5%)	102 (23.2%)
Hypoglycemia on admission	194 (28.2%)	164 (32.3%)	30 (16.7%)	122 (49.2%)	72 (16.4%)
Hypoxia on admission	48 (6.7%)	43 (8.5%)	0 (0%)	43 (8.5%)	0 (0%)
Pupils bluntness on admission	492 (71.5%)	402 (79.1%)	90 (50%)	224 (90.3%)	268 (60.9%)
Neurological deterioration	318 (44.5%)	290 (57.1%)	12 (6.7%)	242 (97.6%)	60 (13.6%)
Seizure	96 (14.0%)	90 (17.7%)	6 (3.3%)	54 (21.8%)	42 (9.5%)
Systolic arterial pressure, mm Hg	130.54 ± 25.50	130.76 ± 27.96	129.93 ± 16.76	133.21 ± 34.73	129.04 ± 18.23
Diastolic arterial pressure, mm Hg	76.85 ± 15.53	76.61 ± 16.46	77.50 ± 12.58	79.09 ± 19.76	75.58 ± 12.39
Mean arterial pressure, mm Hg	95.12 ± 17.93	94.95 ± 19.44	95.60 ± 12.76	97.31 ± 23.76	93.88 ± 13.45
Heart rate, beats/min	85.7 ± 25.2	72.52 ± 0.52	75.55 ± 0.55	78.85 ± 0.62	79.65 ± 0.36
Body temperature, °C	36.86 ± 0.68	36.83 ± 0.72	36.92 ± 0.53	36.88 ± 0.88	36.85 ± 0.53
Blood oxygen saturation, %	95.35 ± 4.37	94.95 ± 19.44	97.27 ± 13.93	92.40 ± 5.53	97.10 ± 2.15
Blood glucose level, mmol/L	9.75 ± 3.41	10.05 ± 3.40	8.93 ± 3.33	11.19 ± 3.60	8.95 ± 3.02
CT scan					
CT classification 5 or 6	431 (62.6%)	329 (65.4%)	102 (56.7%)	216 (81.8%)	240 (54.1%)
Abnormal cisterns on initial CT scan	533 (77.5%)	425 (83.7%)	108 (60.0%)	230 (92.7%)	303 (68.9%)
Midline shift >5 mm on initial CT scan	497 (72.2%)	371 (73.0%)	126 (70.0%)	200 (80.6%)	297 (67.5%)
Presence of traumatic SAH on initial CT scan	592 (86.0%)	442 (87.0%)	150 (83.3%)	230 (92.7%)	362 (82.3%)
Medication treatment					
Mechanical ventilation	260 (37.8%)	260 (51.2%)	0 (0%)	230 (92.7%)	30 (6.8%)
Intracranial surgery in first 24 h	517 (75.1%)	409 (80.5%)	108 (60.0%)	218 (87.9%)	299 (68.0%)
Laboratory					
Leukocyte, 10 ⁹ /L	21.87 ± 3.02	20.44 ± 3.01	19.36 ± 3.25	22.85 ± 2.54	18.60 ± 2.69
Neutrophil, 10 ⁹ /L	17.12 ± 3.21	19.98 ± 2.66	17.96 ± 2.69	19.90 ± 1.69	16.40 ± 1.98
Lymphocyte, 10 ⁹ /L	1.49 ± 0.22	1.40 ± 0.12	1.55 ± 0.23	1.78 ± 0.25	1.31 ± 0.55
NLR	15.64 ± 7.29	15.07 ± 6.63	11.60 ± 4.05	18.75 ± 7.76	13.75 ± 6.27

Abbreviations: CI, confidence intervals; CT, computerized tomography; GCS, Glasgow Coma Scale; NLR, neutrophil to lymphocyte ratio; SAH, subarachnoid hemorrhage.

^aNumerical variables were presented as mean ± standard deviation and categorical variables were expressed as counts (percentage). Difference between groups was compared using the 2-sample t-test for numerical variables and the Pearson χ^2 test or Fisher exact test for categorical variables.^b $P < .0$.

1.064–1.138; both $P < .001$) (see Table 2). ROC curve analysis of the NLR for assessing the predicted performance at 1-year unfavorable outcome was represented as area under the curve (AUC) = 0.723 (95% CI = 0.682–0.763) and the predicted performance sensitivity and specificity were derived as 60.2% and 71.1%, respectively, based on the best threshold of 13.05 for the NLR (see Figure 2).

The predictive performance of the NLR was similar to GCS scores in severe TBI (AUC = 0.719, 95% CI = 0.673–0.766, $P = .9038$) (see Figure 2).

Mortality prediction

A total of 248 patients (36.0%) died from head trauma by 1 year. Similarly as with 1-year unfavorable outcome, higher NLR was associated with 1-year mortality (18.75 ± 7.76 vs 13.75 ± 6.27 , $P < .001$). Four variables—temperature, the NLR on admission, deterioration, and mechanical ventilation—that were significantly associated with 1-year mortality were put into multivariate model analysis. The results showed that decreased temperature, increasing NLR, deterioration, and not using mechanical ventilation were associated with 1-year mortality (temperature: OR = 0.411, 95% CI = 0.281–0.600; NLR: OR = 1.158, 95% CI = 1.094–1.226; deterioration: OR = 334.21, 95% CI = 77.25–1445.9; mechanical ventilation: OR = 0.006, 95% CI = 0.002–0.018; all $P < .001$) (see Table 3). ROC curve analysis for the NLR to assess the predicted performance for 1-year mortality was represented as AUC = 0.704 (95% CI = 0.66–0.744), and the predicted performance sensitivity and specificity were derived as 60.1% and 72.5%, respectively, based on the best threshold of 16.23 for the NLR (see Figure 3). The predictive performance of the NLR was worse than that of GCS scores (AUC for GCS = 0.790, 95% CI, 0.757–0.823, $P = .001$) (see Figure 3).

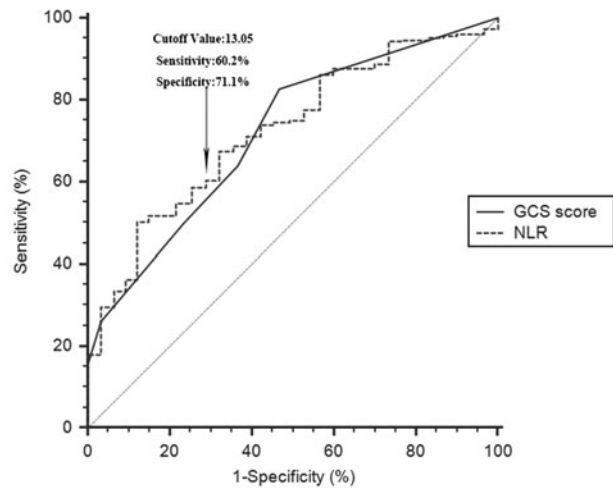


Figure 2. A comparison of AUC for identifying 1-year unfavorable outcome between the GCS and NLR. ROC curves were constructed on the basis of the sensitivity and specificity of the NLR and GCS score for identifying 1-year unfavorable outcome. AUC was calculated on the basis of ROC curves and expressed as 95% confidence interval. AUC ranges from 0.5 to 1.0. An AUC closer to 1 indicates a higher predictive power. AUC indicates area under the curve; GCS, Glasgow Coma Scale; NLR, neutrophil to lymphocyte ratio; ROC, receiver operating characteristic.

DISCUSSION

The current study demonstrated that the values of the NLR on admission of those patients with severe TBI who died or had an unfavorable outcome by 1 year after head trauma were significantly higher than for those patients with severe TBI who survived and had a favorable outcome. Furthermore, in multivariate logistic regression models of predictors of death and unfavorable outcome, the NLR on admission was a significant independent predictor of functional outcome and mortality. Importantly, the prognostic performance of the

TABLE 2 Multivariate model analysis of unfavorable outcomes with predictors^a

Components of models	Multivariate model analysis Odds ratio (95% CI)	Model summary		
		Overall model significance level	Cox and Snell Pseudo- R^2	Nagelkerke Pseudo- R^2
Model I				
Age, y	1.064 (1.05–1.07) ^c	<.001 ^b	12.80%	18.70%
Model II				
Age, y	1.068 (1.052–1.083) ^c	<.001 ^b	17.60%	25.70%
NLR	1.100 (1.064–1.138) ^c			

Abbreviations: CI, confidence interval; NLR, neutrophil to lymphocyte ratio.

^aMultivariate logistic regression analysis was performed with a forward selection (method = Wald).

^b $P < .001$.

^c $P < .001$.

TABLE 3 *Multivariate model analysis of mortality with predictors^a*

Components of models	Multivariate model analysis Odds ratio (95% CI)	Model summary		
		Overall model significance level	Cox and Snell Pseudo- R^2	Nagelkerke Pseudo- R^2
Model I		<.001 ^b	56.00%	76.70%
Mechanical ventilation	0.006 (0.003–0.010) ^c			
Model II		<.001 ^b	62.10%	85.20%
Deterioration	55.989 (21.903–143.118) ^c			
Mechanical ventilation	0.025 (0.012–0.050) ^c			
Model III		<.001 ^b	63.80%	87.50%
NLR	1.141 (1.085–1.200) ^c			
Deterioration	168.68 (48.373–588.259) ^c			
Mechanical ventilation	0.012 (0.005–0.030) ^c			
Model IV		<.001 ^b	64.80%	88.90%
Temperature, °C	0.411 (0.281–0.600) ^c			
NLR	1.158 (1.094–1.226) ^c			
Deterioration	334.209 (77.245–1445.98) ^c			
Mechanical ventilation	0.006 (0.002–0.018) ^c			

Abbreviations: CI, confidence interval; NLR, neutrophil to lymphocyte ratio.

^aMultivariate logistic regression analysis was performed with a forward selection (method = Wald).

^b $P < 0.001$.

^c $P < .001$.

NLR for functional outcome was similar to that of the GCS score, which substantiates the potential of the NLR as a new prognostic marker in severe TBI. However, the

predictive performance of the NLR for mortality was significantly lower than that of the GCS score, which suggests that the value of the NLR as a prognostic marker in patients with severe TBI might be more useful for predicting functional outcome than for predicting mortality.

Severe TBI is associated with high morbidity and mortality. Early prognostication of the risk of death or a long-term poor outcome would facilitate optimized care and improved allocation of healthcare resources. The GCS score is well known to be associated with mortality and poor clinical outcomes of TBI. However, finding a readily measurable marker for predicting clinical outcomes in patients with severe TBI would be helpful for early prognostication and risk stratification. The NLR, derived from the white blood cell differential count, a routine laboratory study, is easy to obtain and calculate, and easy to integrate into daily practice, and does not add extra costs. Another advantage of the NLR is that it is more objective than the GCS score. Furthermore, slight changes in a patient's physical condition may not be reflected in the GCS score whereas it may be reflected in the NLR, which could lead to changes in treatment. One drawback of the NLR is the average turnaround time for laboratory results for the NLR in the emergency department, which is between 30 and 40 minutes. In comparison, head CT images are available to be viewed within 10 minutes and the GCS score can be immediately obtained from physical examination of the patient.

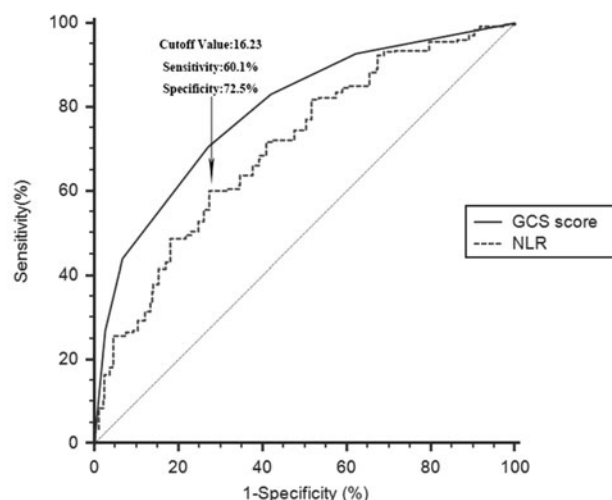


Figure 3. A comparison of AUC for identifying 1-year mortality between the GCS and NLR. ROC curves were constructed on the basis of the sensitivity and specificity of the NLR and GCS score for identifying 1-year mortality. AUC was calculated on the basis of ROC curves and expressed as 95% confidence interval. AUC ranges from 0.5 to 1.0. An AUC closer to 1 indicates a higher predictive power. AUC indicates area under the curve; GCS, Glasgow Coma Scale; NLR, neutrophil to lymphocyte ratio; ROC, receiver operating characteristic.

Our study only included patients with severe TBI; there were no patients with mild or moderate TBI. Thus, the findings on the prognostic value of the NLR for patients with mild or moderate TBI is not yet clear, which is a limitation of this study. The prognostic value of the NLR in mild and moderate TBI warrants further investigation.

CONCLUSION

In this study, increased NLR at admission of patients with severe TBI was associated with unfavorable 1-year functional outcome and mortality. The NLR may have potential as an easily obtained clinical marker for predicting outcome in patients with severe TBI.

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